RESEARCH LETTER

Trends in County-Level MMR Vaccination Coverage in Children in the United States

Childhood vaccination has substantially reduced the incidence of vaccine-preventable diseases and saved countless children's lives worldwide.[1] The US childhood vaccination program is estimated to have prevented more than 24 million cases of vaccine-preventable diseases in 2019 alone.[2] Nonetheless, there is evidence of a US national-level decline in the childhood measles-mumps-rubella (MMR) vaccination rate between 2019 and 2024 (**Figure 1**),[3] and the Centers for Disease Control and Prevention (CDC) has reported a resurgence of measles outbreaks,[3] a sentinel indicator of childhood vaccine interruptions.[4] As of May 14, 2025, 1001 confirmed measles cases have been reported by 31 jurisdictions (the most cases reported in the US in a single year in >3 decades, with the exception of 2019), with most occurring in unvaccinated children.[3] If vaccination rates continue to decline, measles is likely to return to endemic levels in the US.[5]

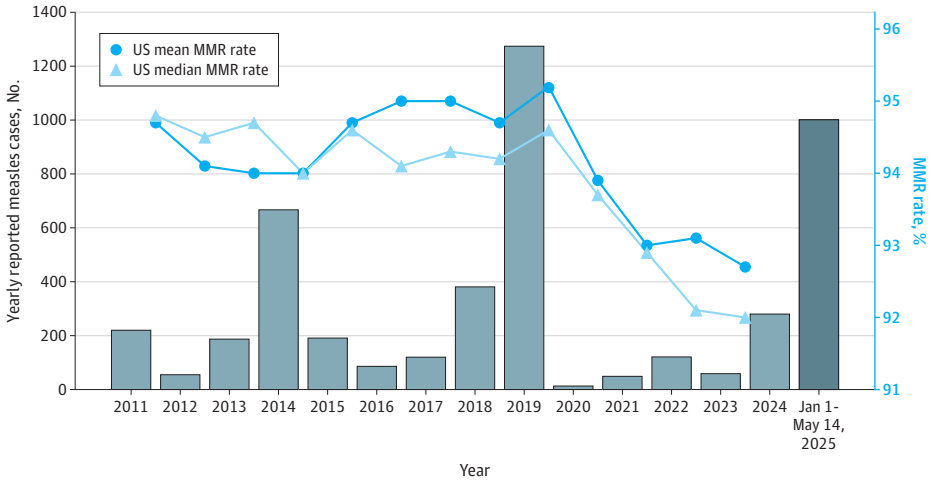
Supplemental content

Although the national- and state-level declines in MMR coverage are well documented, MMR vaccination coverage can vary substantially within a state.[6] We generated a standardized dataset with annual county-level vaccination rates for children from 2017 to 2024 for all states in the US where this information was available and evaluated spatiotemporal trends in vaccination coverage during this period. This open, high-resolution dataset serves as a resource to explore the US vaccination landscape and its implications for vaccine-preventable disease.

Methods | We collected county-level 2-dose MMR vaccination rates for kindergarteners from each state's department of health website for each of school years 2017-2018 to 2023-2024. In states where this variable was not available, we used the most comparable proxy, either 2-dose MMR rates for kindergarten through grade 12 students or 5- to 6-year-olds or completion rates for a combined vaccine series that included 2-dose MMR. A detailed description of the data sources and collection process, including discrepancies in reporting by state, is in the eAppendix in Supplement 1 and eTable in Supplement 2. Institutional review board approval was not required because data were publicly available and deidentified. Data were analyzed with Python version 3.13.3 (Python) and pandas version 2.2.3 (Python).

Results | The final dataset included at least 1 year of vaccination data for 2237 counties across 38 states, of which 23 states report 2-dose MMR data for kindergartners, 4 states report 2-dose MMR data for a wider age range, and 11 states report 2-dose MMR coverage as part of a full vaccination series for school-age children. **Figure 2**A and B illustrate the county-level 2-dose MMR vaccination (or full vaccination series) rates before and after the pandemic, respectively; Figure 2C illustrates the distribution of county-level vaccination rates for the prepandemic and postpandemic periods. These data reveal a trending nationwide decline in vaccination rates during the COVID-19 pandemic, with significant heterogeneity within and across states. Across the 2066 counties in 33 states with both prepandemic and postpandemic vaccination rates, the county-level mean (SD) vaccination rate decreased from 93.92% (5.71%) prepandemic to 91.26% (6.95%) postpandemic, a mean decline of 2.67% (5.27%). Of these counties, 1614 of 2066 (78%) reported a decline in coverage during

Figure 1. Reported Measles Cases and Measles-Mumps-Rubella (MMR) Vaccination Rates in the US



Reported measles cases from 2011 to 2025 and national mean and median MMR vaccination rates in the US from school years 2011-2012 to 2023-2024. The 2025 cases include only those reported between January 1, 2025, and May 14, 2025. (Source: Centers for Disease Control and Prevention[3] as of May 14, 2025.)

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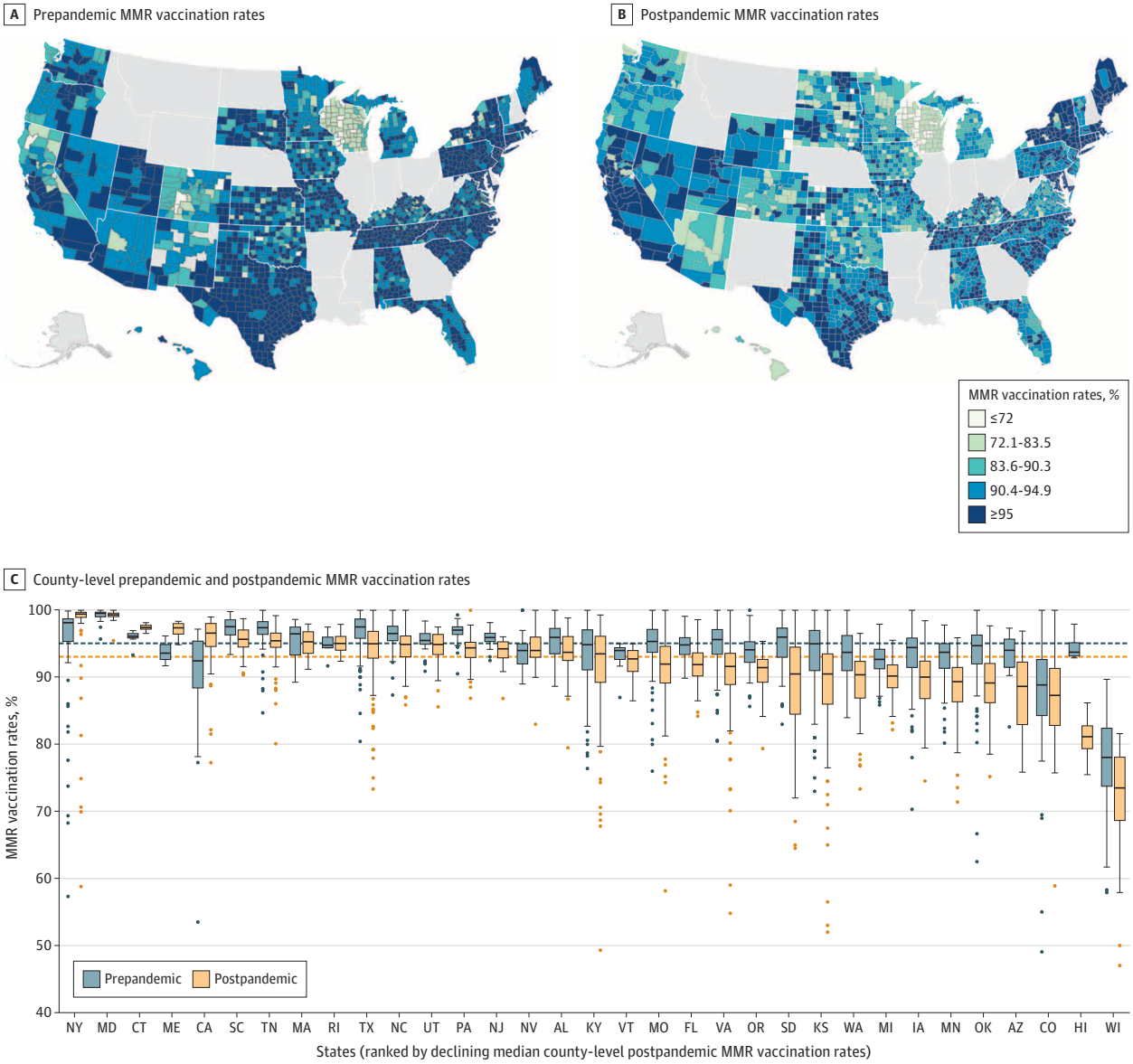
Figure 2. County-Level Measles-Mumps-Rubella (MMR) Vaccination or Complete Immunization Series Rates



Maps plotting the county-level 2-dose MMR vaccination (or complete immunization series) rates before (A) and after (B) the COVID-19 pandemic for states with data available. States and counties in gray have missing vaccination data. C, Box plot showing the county-level 2-dose MMR vaccination (or complete immunization series) rates for each state before (gray) and after (orange) the pandemic for the 33 states with vaccination data available in both periods. For each county, the prepandemic vaccination rate is the mean of the rates for the 2017-2018, 2018-2019, and 2019-2020 school years, whereas the postpandemic rate is the mean of the rates for the 2022-2023 and 2023-2024 school years for each county (when the data were available). Each box plot shows the distribution of the vaccination rates for all counties in a state for the specified period. The annual national mean of 2-dose kindergarten MMR vaccination rates reported by the Centers for Disease Control and Prevention was used to compute the prepandemic and postpandemic national means of 95% (gray dashed line) and 93% (orange dashed line), respectively.[3] Maps were created with ArcGIS Pro version 3.4 (Esri).

this period. Only 4 of the 33 states in our study—California, Connecticut, Maine, and New York—reported an increase in the median county-level vaccination rate.

Discussion | Our county-level dataset complements the state- and national-level CDC data,[3] confirming a widespread decline in MMR vaccination rates in the US after the COVID-19 pandemic while revealing significant heterogeneity in vaccination patterns within and across states. This dataset can be used in spatial and statistical analyses to identify factors associated with low or declining MMR rates in US counties and help inform targeted vaccination strategies to reduce the risk of measles outbreaks.

This study has limitations. Inconsistencies and gaps in reporting across states restricted analysis to 33 states and affected comparability of the data. Reported vaccination data may

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be inaccurate due to missing student records, home-schooled children, and select schools that are not required to provide records, thus potentially introducing bias.

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Author Contributions: Drs Dong and Gardner had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.
Concept and design: Dong, Gardner.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: Dong, Saiyed, Gardner.
Critical review of the manuscript for important intellectual content: All authors.
Statistical analysis: Dong, Saiyed.
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Health Care Workforce Recovery After the End of the COVID-19 Emergency

The COVID-19 pandemic caused drastic changes in health care utilization, delivery, and clinical organization finance.[1] Initial research showed that health care employment did not fully recover by 2022's end, with trends varying by setting.[2,3] Employment in physician offices reached full recovery, while long-term care facility staffing was 10.5% below prepandemic levels in 2020-2022.[3] The intensive behavioral health services workforce was slow to recover, despite increased demand for these services during the pandemic.[4] Although the COVID-19 public health emergency expired in May 2023, the pandemic's full impact on the health care workforce and its subsectors remains unclear.

Supplemental content

Methods | This study used industry- and national-level employment data from the US Bureau of Labor Statistics' 2016-2024 Quarterly Census of Employment and Wages to assess health care employment changes relative to changes in non-health care sectors from 2016-2024 (eAppendix in Supplement 1). Analyses also evaluated key health care subsectors, including physician offices, hospitals, skilled nursing facilities (SNFs), office-based behavioral health practitioners, and intensive behavioral health facilities.

Ordinary least-squares regressions with seasonality adjustments were estimated using data in the prepandemic period (quarter 1 of 2016 to quarter 4 of 2019) to generate predicted employment levels in the postpandemic period (quarter 1 of 2020 to quarter 3 of 2024). This exercise simulated what employment would have been in the absence of the pandemic, assuming prepandemic trends continued. Separately, a model was estimated that adjusted for the quarterly civilian labor force to account for US adult population and labor force growth. Analyses were conducted with Stata version 18.0 (StataCorp). This study was granted not-regulated status by the University of Michigan Medical School Institutional Review Board and followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for cross-sectional studies. Because data were publicly available and deidentified, this study was exempted from human subjects review.

Results | Between quarter 4 of 2019 and quarter 2 of 2020, health care employment decreased by 6.9% relative to predicted levels in the absence of the pandemic, from 22 456 615 to 21 058 286 jobs (**Figure**, A, and **Table**). Subsequently, aggregated health care employment recovered, increasing to 24 401 427 jobs in quarter 3 of 2024 (0.2% below predicted levels). In contrast, non-health care employment totaled 127 548 688 jobs in quarter 4 of 2019, with a larger decline in quarter 2 of 2020 (11.4% below predicted levels). Non-health care employment recovered more slowly than health care employment in quarter 3 of 2024 (2.9% below predicted levels).

Recovery patterns varied by health care subsectors, with enduring declines in employment in SNFs, hospitals, and intensive behavioral health facilities compared with quicker recoveries among offices of physicians and large increases in employees of office-based behavioral health practitioners (Figure, B-C, and Table). In quarter 2 of 2020, employment in health care subsectors decreased by between 3.4% and 7.6% (in hospitals and SNFs, respectively). In quarter 3 of 2024, the differences in observed vs predicted levels were 0.7% for hospitals, −0.1% for SNFs, 1.2% for offices of physicians, −0.8% for intensive behavioral health facilities, and 40.6% for office-based behavioral health practitioners. Employment levels for office-based behavioral health practitioners grew by 84%, from 183 339 in quarter 4 of 2019 to 337 639 in quarter 3 of 2024.

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